Olga Petrovska, SFHEA

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Employment

- 2022 – · · · · **Head of Final Year** at Computer Science Department, Swansea University, UK
  Interim Programme Director in 2022/23 (maternity cover)
- 2020 – · · · · **Teaching Fellow** at Computer Science Department, Swansea University, UK
- 2020 – 2025 **Technology Consultant** and **Project Manager** at Ukraïner.net, remote *(part-time)*
- 2016 – 2020 **Teaching Assistant** at Swansea University, UK *(part-time)*
- 2013 – 2015 **Project Manager / Team Leader** at Wolfestone, Swansea, UK
- 2010 – 2012 **Web Production Lead / Internal Relationships Leader** at IBM, Bratislava, Slovakia

Positions of Responsibility and Community Service

- 2025 – · · · · **Leader** of the "Embracing AI in Computer Science" Working Group within Swansea University's CoSTA (Computer Science Teaching Attainment Group)
- **Co-leader** of the ITiCSE Working Group on Refactoring Upper Computing Courses to Leverage GenAI
- **Program Committee Member** & **Poster Track Chair** at Human-Centered AI Education & Practice Conference (HCAI-ep'26),
- **Program Committee Member** at Computing Education Practice (CEP 2026), and Continuity, Computability, Constructivity (CCC 2025)
- **Reviewer** of HEA fellowship applications under Swansea University's Advance HE-accredited Celebration of Professional Recognition (CoPR) programme
- 2024 – · · · · **Reviewer** for the several international journals (Frontiers in Education, Information and Software Technology, Media and Communication, and IEEE Access) and conferences (SIGCSE TS 2026, OpenEd25, Koli Calling'25, and BCSWomen Lovelace Colloquium)
- 2023 – · · · · **Council Member** at the Association of Computability in Europe
- **Project Support Officer** at the UK-Ukraine Academic Mentorship Scheme
- 2020 – · · · · **Treasurer** at the British Colloquium for Theoretical Computer Science

Education

- 2016 – 2021 **Ph.D., Computer Science**, Swansea University, UK
  Thesis: *Enhanced Realizability Interpretation for Program Extraction*
- 2015 – 2016 **M.Sc., Computer Science**, Swansea University, UK
  Thesis: *Prawf: Interactive Proof Assistant for Predicate Logic*
- 2006 – 2007 **Master's Degree, Linguistics**, Chernivtsi National University, Ukraine
  Thesis: *The Functional-Semantic Field of Quantity in John Fowles' Discourse*
- 2002 – 2006 **Bachelor's Degree, Pedagogy**, Ternopil National Pedagogical University, Ukraine

CPD Courses and Certifications

- 2025 — **Personal Development Course for External Examiners**, Advance HE
- **Level 4 NVQ Diploma in Management**, ILM
- **Project Management Professional Certificate**, Google
- **Sustainability in the Digital Age: Energy-Efficient Software Development**, openHPI
- **Sustainability in the Digital Age: Environmental Impacts of AI Systems**, openHPI
- 2024 — **AI in Education: Leveraging ChatGPT for Teaching**, University of Pennsylvania
- 2023 — **Data Science Bootcamp**, openHPI
- 2022 — **Introduction to Quantum Computing with Qiskit (with IBM Quantum)**, openHPI
- **Sustainable Software Engineering**, openHPI
- **Cyberthreats by Malware**, openHPI
- **Digital Identities**, openHPI
- **Confidential Communication in the Internet**, openHPI

Teaching Experience and Course Design

- 2025 – · · · · — Delivery of **CSB120MC Software Testing**
- Design and delivery of **CSF1007 Professional Issues**
- 2021 – · · · · — Design and delivery of **CSF302 Project Planning and Management**
- Design and delivery of **CSF206 Algorithms and Automata**
- Coordination of final year project modules: **CSF301 Project Specification and Development** and **CSF300 Project Implementation and Dissertation**
- Module support for **CS-170/CS-175 Modelling Computing Systems**
- 2024 – 2025 — Co-designed **CSF325 Artificial Intelligence**
- 2021 – 2025 — Designed and delivered **CSF107 Professional Issues of Software Engineering**
- Designed and delivered a range of short courses and workshops for teachers, college students, and general public on **Artificial Intelligence, Cyber Security, Graphic Design, Web Development, Excel**, and other.
- 2022 – 2023 — Delivered **CSB100MC Computational Thinking**
- Delivered **CSB300MC Software Engineering Project Management**
- Delivered components of **CS-175 Modelling Computing Systems 2**
- Delivered components of **CSC301/CSCM01 Software Engineering Project Planning and Management**
- 2021– 2022 — Delivered components of **CSF399 Human-Computer Interaction**
- 2020 – 2021 — Delivered components of **CSF207 Computer Security**
- Delivered **CSF106 Discrete Mathematics for Computer Science**
- 2016 – 2020 — Provided support with labs, tutorials and assessment in a variety of CS modules:
  - **CS-205 Declarative Programming**
  - **CSCM13 Critical System / CSC313 High Integrity Systems**
  - **CSCM75 Logic in Computer Science**
  - **CS-275 Automata Theory and Formal Languages**
  - **CS-081 Introduction to Algorithms and Data Structures**

Publications

Conference Proceedings

1 O. Andrei, S. W. Nabi, M. Barr, and **O. Petrovska**, "Integrating socially responsible computing competencies in computer science and software engineering education," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, pp. 36–37, ISBN: 9798400711725. DOI: 10.1145/3702212.3702225.

2 D. J. Bouvier, B. P. Cipriano, R. Glassey, R. Pettit, E. Anderson, A. Birillo, R. Dougherty, O. Hazzan, **O. Petrovska**, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, A. Vadaparty, and G. R. Yeluripati, "GenAI integration in upper-level computing courses," in *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2*, ser. ITiCSE 2025, Nijmegen, Netherlands: Association for Computing Machinery, Jun. 2025, pp. 691–692, ISBN: 9798400715693. DOI: 10.1145/3724389.3731276.

3 L. Clift and **O. Petrovska**, "Learning without limits: Analysing the usage of generative ai in a summative assessment," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, pp. 5–8, ISBN: 9798400711725. DOI: 10.1145/3702212.3702214.

4 **O. Petrovska**, L. Clift, and F. Pantekis, "Video in assessments for soft skill development and evaluation," in *Proceedings of the 2025 Conference on UK and Ireland Computing Education Research*, ser. UKICER'25, Association for Computing Machinery, Sep. 2025, ISBN: 9798400720789. DOI: 10.1145/3754508.3754531.

5 **O. Petrovska**, R. Pearsall, and L. Clift, "Assessing software engineering students' analytical skills in the era of generative ai," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, p. 34, ISBN: 9798400711725. DOI: 10.1145/3702212.3702223.

6 **O. Petrovska**, L. Clift, F. Moller, and R. Pearsall, "Incorporating generative AI into software development education," in *Proceedings of the 8th Conference on Computing Education Practice*, ser. CEP'24, Durham, United Kingdom: Association for Computing Machinery, Jan. 2024, pp. 37–40, ISBN: 9798400709326. DOI: 10.1145/3633053.3633057.

7 **O. Petrovska**, L. Clift, and F. Moller, "Generative AI in software development education: Insights from a degree apprenticeship programme," in *Proceedings of the 2023 Conference on United Kingdom & Ireland Computing Education Research*, ser. UKICER'23, Swansea, Wales UK: Association for Computing Machinery, Sep. 2023, ISBN: 9798400708763. DOI: 10.1145/3610969.3611132.

8 U. Berger, **O. Petrovska**, and H. Tsuiki, "Prawf: An interactive proof system for program extraction," in *Beyond the Horizon of Computability*, M. Anselmo, G. Della Vedova, F. Manea, and A. Pauly, Eds., Cham: Springer International Publishing, 2020, pp. 137–148, ISBN: 978-3-030-51466-2. DOI: 10.1007/978-3-030-51466-2_12.

9 U. Berger and **O. Petrovska**, "Optimized program extraction for induction and coinduction," in *Sailing Routes in the World of Computation*, F. Manea, R. G. Miller, and D. Nowotka, Eds., Cham: Springer International Publishing, 2018, pp. 70–80, ISBN: 978-3-319-94418-0. DOI: 10.1007/978-3-319-94418-0_7.

Other

1 **O. Petrovska**, *Course Portfolio on Algorithms and Automata (Apprenticeship)*, 2024. DOI: 10.5281/zenodo.14448266.

2 **O. Petrovska**, *Enhanced Realizability Interpretation for Program Extraction*, 2021. DOI: 10.23889/suthesis.57831.

Selected Talks

- **Modular Generative AI Training to Promote AI Literacy Across the University** at the Swansea University annual Learning and Teaching Conference SUSALT 2025 - Enhancing Our Practice: Share - Learn - Create | co-presented with Filippos Pantekis (10.07.2025, Swansea University, Wales)
- **The Art of Teaching Theory Across Diverse Backgrounds** at the British Colloquium for Theoretical Computer Science 2025 (14-16.04.2025, University of Strathclyde, Scotland)
- **GenAI in Education and Research** at the NAIADES (Network for Artificial Intelligence, the Arts, the Digital Economy and Society) Workshop "GenAI: Use and Best Practices" (20.02.2025, Swansea University, Wales)
- **Computer Science Outreach in Higher Education: Challenges and Opportunities** | co-presented with Megan Venn-Wycherley (08.01.2025, Durham University, England)
- **Developing Students' Critical Skills in the AI-enabled World** at the Advance HE Assessment and Feedback Symposium 2024 (05.11.2024, online)
- **How are students using Generative Artificial Intelligence?** (01.11.2024, "Pinch of SALT" podcast)
- **Generative AI and Conversations with Your Students** at the Lunch and Learn Webinar of the Swansea Academy of Learning and Teaching | co-presented with Lee Clift (22.10.2024, online)
- **Student vs AI** at the Learning and Teaching Enhancement Centre (LTEC) Conference "GenAI: Challenges and Solutions" (18.09.2024, Swansea University, Wales)
- **Operational Semantics in Prawf** at the Workshop on Proof and Computation 2022 (01.06.2022, Kloster Schlehdorf, Germany)
- **Intuitionistic Fixed Point Logic and Program Extraction** at the Workshop on Continuity, Computability, Constructivity 2018 (24-28.09.2018, University of Algarve, Faro, Portugal)
- **Natural Language Proofs for Program Extraction** at the International Workshop on Proof, Computation, Complexity 2017 (26-27.07.2017, University of Göttingen, Germany)
- **Natural Language Proof Checking in Predicate Logic** at the Workshop on Proofs, Programs, and Verification (19.04.2017, University of Canterbury, Christchurch, New Zealand)

Event Organisation

Sep 2025	**CCC 2025** (Continuity, Computability, Constructivity) – Local organiser[1]
Mar 2025	**UK-Ukraine Research Twinning Showcase Event 2025** at the Liverpool University – Co-organiser
Jul 2024	**CCA 2024** (International Conference on Computability and Complexity in Analysis) – Local organiser
May 2024	**Agda Implementors' Meeting XXXVIII** – Local organiser
Sep 2023	**UKICER 2023** (UK and Ireland Computing Education Research) – Local organiser
Jul 2022	**CiE 2022** (Computability in Europe: Revolutions and Revelations in Computability) – Local organiser
Apr 2022	**BCTCS 38** (British Colloquium for Theoretical Computer Science) – Co-chair
Apr 2020	**BCTCS 36** online conference – Local organiser

[1] All conferences/events organised at Swansea University, unless stated otherwise.

Event Organisation (continued)

Sep 2019 — **2nd International Summer School on Proof Theory** and **Workshop on Proof Theory and its Applications** – Local organiser

Secondments and Residences

- Jul-Aug. 2018 — **Residence** at the Hausdorff Trimester Program (Hausdorff Research Institute for Mathematics, Germany)
- Mar-Apr. 2018 — **Secondment** at the Mathematical and Information Sciences Division of the Human and Environmental Studies Faculty (Kyoto University, Japan)
- Mar-May. 2017 — **Secondment** at the School of Mathematics and Statistics (University of Canterbury, New Zealand)

Languages

- Native or bilingual proficiency — English, Ukrainian, Slovak, Russian
- Limited working proficiency — German, Polish
- Elementary proficiency — Welsh, Mandarin Chinese